

Electrical Single-Line Diagram & Specifications

Doc No: ELEC-002
Rev: B
Date: 2026-05-10

Project: Lakefront Office Tower — Phase II

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SECTION 26 05 53 — ELECTRICAL IDENTIFICATION & SINGLE-LINE

1.0 SYSTEM OVERVIEW

The electrical distribution system for Lakefront Office Tower Phase II is a 15 kV utility service stepped down through two 2,000 kVA pad-mounted transformers (XFMR-1 and XFMR-2) to a 480/277V, 4-wire grounded-wye main distribution panel (MDP) in the basement electrical room. A 750 kW diesel generator provides emergency standby service. See drawing E1.01 for complete single-line diagram.

2.0 SINGLE-LINE DIAGRAM

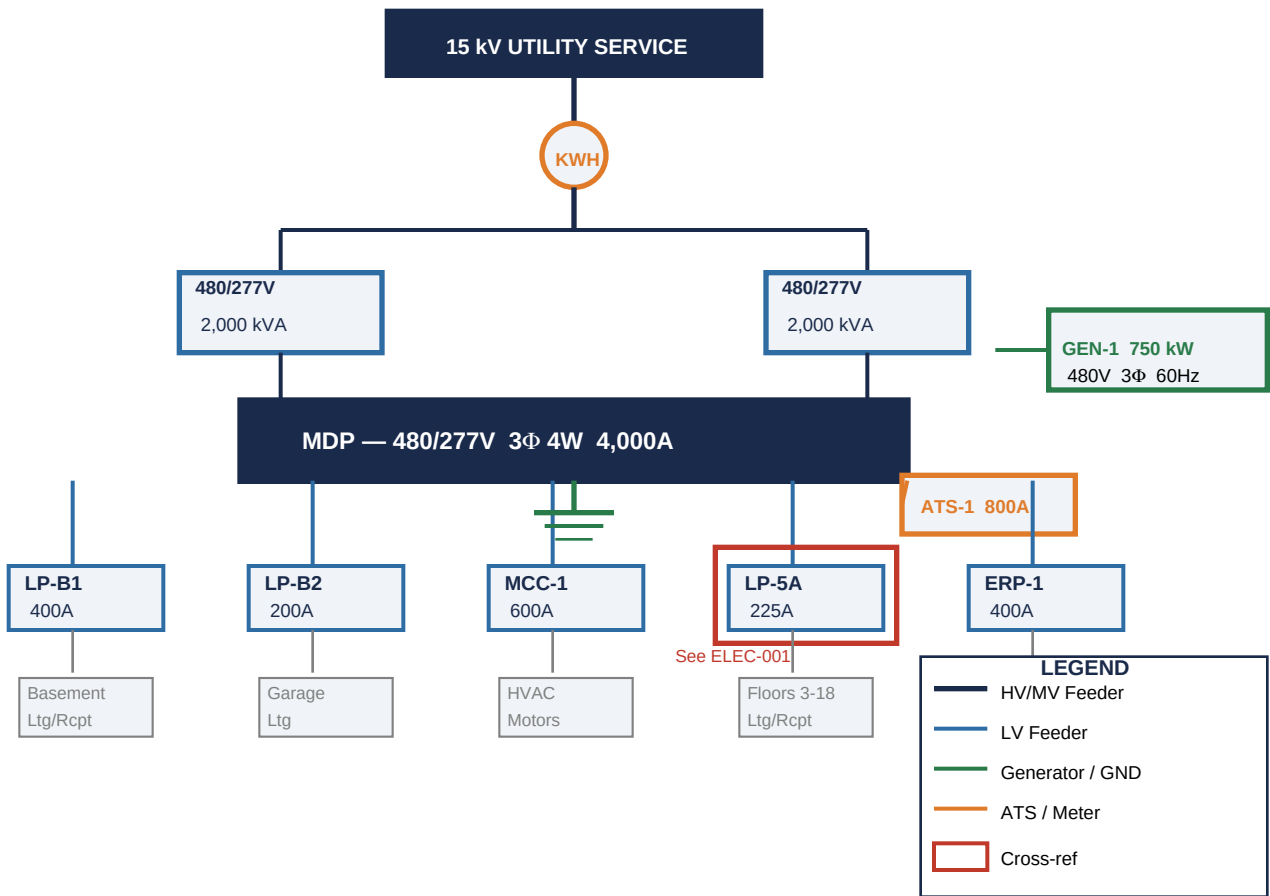


Figure 2 — Electrical Single-Line Diagram (Schematic — Not to Scale)

3.0 ELECTRICAL EQUIPMENT SCHEDULE

Tag	Description	Rating	Manufacturer / Model	AIC
XFMR-1/2	Pad-Mounted Transformer	2,000 kVA, 15kV–480Y/277V	ABB / OFAF-2000	—
MDP	Main Distribution Panel	4,000A, 480/277V 3Φ 4W	Square D / I-Line	65 kAIC
ATS-1	Automatic Transfer Switch	800A, 480V	Asco / 7000 Series	—
GEN-1	Emergency Generator	750 kW, 480V 3Φ	Caterpillar / 3512C	—
LP-5A	Lighting Panel Floor 5	225A, 208Y/120V 3Φ 4W	Square D / QO	22 kAIC
MCC-1	Motor Control Center	600A, 480V, NEMA 12	Allen-Bradley	65 kAIC
ERP-1	Emergency Panel	400A, 480/277V	Square D / I-Line	42 kAIC

4.0 INSTALLATION NOTES

- All work shall comply with NEC 2023, NFPA 70E, local ordinances, and utility company requirements.
- Coordinate service entrance with ComEd prior to construction. Submit utility applications 90 days minimum before energization.
- Emergency system wiring (ERP-1) shall be in dedicated conduit, separate from normal power.
- ATS shall have closed-transition transfer capability. Generator exerciser timer: weekly, 30-min loaded test.
- All MV cables: 15 kV-rated EPR insulation, shielded, in rigid steel conduit below grade.
- Provide fault current study and arc-flash hazard analysis prior to energization per NFPA 70E.
- Label all equipment with arc-flash warning per NFPA 70E. Incident energy analysis required.

END OF SECTION 26 05 53